SJÄLVSTÄNDIGA ARBETEN I MATEMATIK

MATEMATISKA INSTITUTIONEN, STOCKHOLMS UNIVERSITET

Gröbner Bases and Elimination in Macaulay 2

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Abstract

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1 Introduction

Hello world
[DAC10]
[LM21]

2 The Algebreic Abstraction of Geometry

Algebra is the mathematical expression of the human need for structure and abstraction. Through algebra, humans try to capture more of the mathematical, and natural, domain under their pen. The endless hunger for knowledge can express itself in various way. When using algebra as a tool, creating abstract models that capture generic structure is favored. However, these abstract models can be difficult to understand. The ability to navigate between levels of abstraction is a vital tool for any mathematician interested in applying algebreic methods.

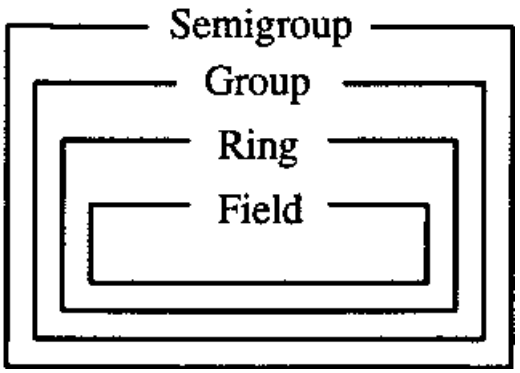
This paper assumes a degree of familiarity with algebreic structures, but the uninitiated peruser is kindly directed towards the easy-going introduction of "Introducetion to Modern Algebra" [Wei70] with bite size exercises, or alternativly, the more rigorous "Abstract Algebra" [DF03] for a deeper understanding.

As the section header suggests, this section will be primarily focused with defining the tools necessary to build an analytic bridge between geometry and algebra so that we can analyze geometric objects with algebra. A function that maps geometric objects to algebreic must be established. This function uses Affine Varietes and Ideals, along with the dictionary established in the Hilbert Strong Nullstellensatz, to accomplish that goal.

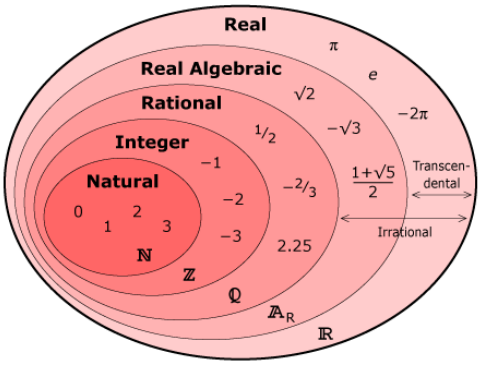
2.1 Levels of Abstraction

Abstract algebra traverses through layers of abstraction as an exercise. As an example, if we are interested in how groups act when supplemented with commutivity (Abelian groups), it would be useful to start at groups and traverse the layers of figure 1a downwards. Note that climbing upwards may not be as fruitful for this particular case.

Choosing an Abelian group that stays beteween groups and rings is $(\mathbb{R}, +)$. Here the set of real numbers is only considered with the binary operator $+$, and commutivity follows. Analyzing this group and all the properties that are associated with it is of interest. However, much use can be extracted from climbing further down the abstraction ladder and analyzing rings a. Keeping it concrete with numbers, one can consider the abelian ring $\mathbb{Z}$. Time can be spent on this layer of abstraction and knowledge can be obtained through leveraging $\mathbb{Z}$'s special properties. Going even further down the ladder, to a field, one could analyze $\mathbb{Q}$. At each layer, new intuition can be gained. Note also, that while traversing downwards in the group abstraction,



(a) Abstractions within Groups [Lat]



(b) Generalization of Numbers [Diand]

Figure 1: Traversing the ladder of Abstraction

we were climbing up and down in the generalization of numbers. Starting at the top with $\mathbb{R}$, we jumped all the way down to $\mathbb{Z}$, and then up again to $\mathbb{Q}$. It is important to consider this, as different layers of abstraction do not neccessarily have to correspond with another. Caution is required when dealing with this iterative abstract process.

2.2 Polynomials

In order to adress the problem at hand fully, we need to generalize the idea of a polynomial. This is done by compressing more and more inforamation into a single mathematical expression. A process of abstraction is necessary. We start with the monomial, continue to a polynomial, and finally consider polynomial rings. All work done is done on these polynomials rings, and this process of abstraction enables us to say something about all monomials simultaneously.

Definition 2.1. A **monomial** in $x_1, \ldots, x_n$ is a product of the form

$$x^\alpha = x_1^{\alpha_1} \cdot x_2^{\alpha_2} \cdots x_n^{\alpha_n} \tag{1}$$

where all $\alpha_1, ..., \alpha_n$ are nonnegative integers.

As the root word "mono" suggests, there is only one of these terms x^α.However, as we are creating the tools to generalize and deal with more abstract concepts, we also want to include the ability to capture any linear combination of these monomials.

Definition 2.2. A **polynomial** f in $x_1, \ldots, x_n$ with coefficients, $a_1, \ldots, a_n$, in a

field $\mathbb{K}$ is a finite linear combination of monomials.

$$f = \sum_{\alpha} a_\alpha x^\alpha, \quad a_\alpha \in \mathbb{K} \tag{2}$$

With the polynomial definitions at hand, we can construct the final abstraction for our mathematics involving polynomials, $\mathbb{K}[x] = \mathbb{K}[x_1, \ldots, x_n]$. An important note is that $\mathbb{K}[x]$ satisfies the axioms for rings and is therefore refered to as the *polynomial ring* When leveraging this powerful abstraction, we capture n dimensions of variables on any field $\mathbb{K}$. This enables more compact analysis.

2.3 Affine Spaces and Varieties

To start the bridge between the world of Algebra as we know it and geometry, there must be an established way to discuss geometry algebreically. This is where the concepts of affine spaces comes in.

This concept allows the analysor to view the geometry algebreically. However, this is not the endgoal, but an analytic step between geometry and the more powerful algebreic tools that we want to apply to geometry.

Affine maps as generalizations of linear maps. [Gal25]

Vector spaces being limiting since origin is given special consideration, bad [Gal25]

Frame invariance.

2.3.1 Affine Spaces

Definition 2.3. An *affine space* is either the empty set, or a triple $< E, \vec{E}, + >$. E is a nonempty set of points while $\vec{E}$ is a vector space (of translations). The action $+$ is an operation on both E and $\vec{E}$ defined as follows: $+ : E \times \vec{E} \to E$. The following axioms are assumed for affine space:

A1 $a + 0 = a$ for every $a \in E$.

A2 $(a + u) + v = a + (u + v)$ for every $a \in E$ and every $u, v \in \vec{E}$.

A3 For any two points $a, b \in E$, there is a unique $u \in \vec{E}$ such that $a + u = b$

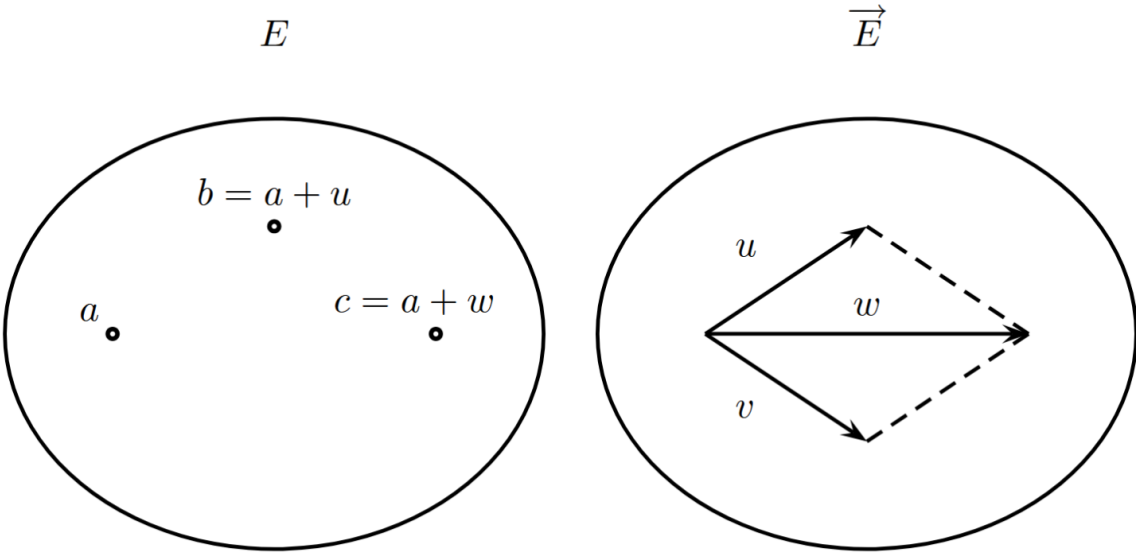


Figure 2: Intuiative understanding of affine space [Gal25]

2.3.2 Affine Varieties

2.4 Rings and Ideals

2.5 Ordering Polynomials

2.6 An Analytic Bridge

2.7 Hilbert Strong Nullstellensatz

3 Gröbner Bases

3.1 Hilbert Bases

3.2 Gröbner Bases

3.3 Properties of Gröbner Bases

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